DO NOW — RECALL

Lesson 133 · Hashing for Integrity, Passwords and Data

Name: _____ Class: _____ Date: _____

1. How does hybrid encryption usually protect bulk web traffic?

2. What are the outputs of a half adder when A=1 and B=1?

3. In Big O notation, what is binary search worst-case complexity?

4. When does a linear search stop?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Use asymmetric establishment/authentication then symmetric traffic keys.

2. Model answer

Sum 0, carry 1.

3. Model answer

O(log n) for sorted random-access data.

4. Model answer

On a match or after examining all items.

By the end, you can…

• A cryptographic hash produces a reproducible fingerprint of file content.

• A matching trusted digest makes accidental/tamper changes less likely.

• Password hashing, integrity checking and hash-table indexing are different applications.

Vocabulary — know these terms

hash function — Maps input data to a fixed-size or bounded-size result.

digest — Output of a cryptographic hash function.

collision — Two distinct inputs produce the same hash result.

salt — Random value combined with a password before secure password hashing.

NEW LEARNING

Read, annotate and explain

Compute a digest

A cryptographic hash produces a reproducible fingerprint of file content.

Compare for integrity

A matching trusted digest makes accidental/tamper changes less likely.

Choose correct use

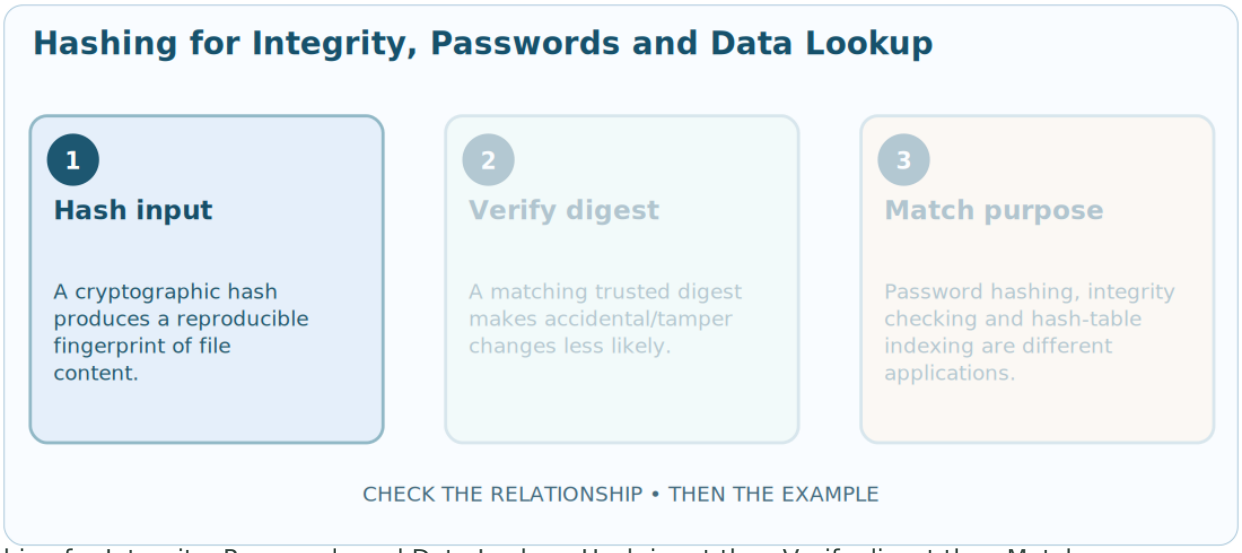
Password hashing, integrity checking and hash-table indexing are different applications.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

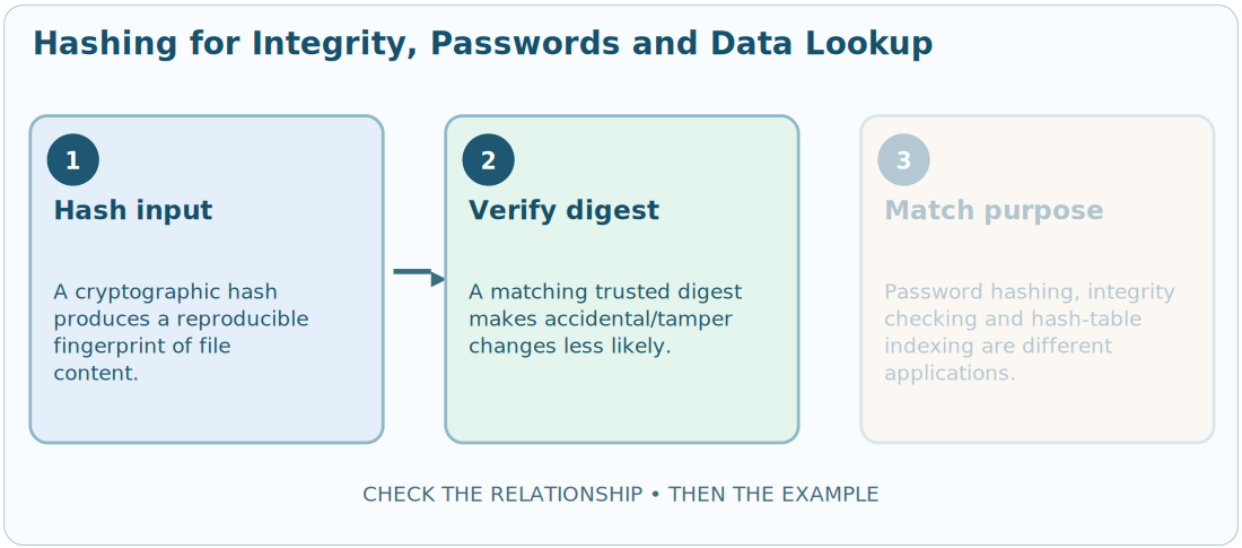
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



Hashing for Integrity, Passwords and Data Lookup; Hash input then Verify digest then Match purpose

New Learning

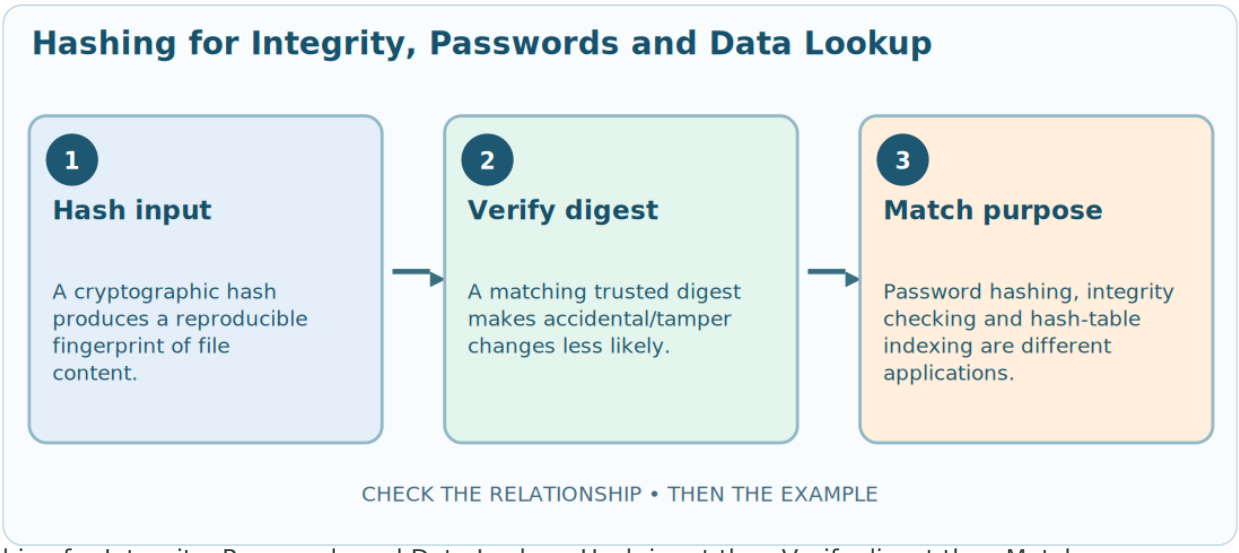


Hashing for Integrity, Passwords and Data Lookup; Hash input then Verify digest then Match purpose

New Learning · original lesson diagrams

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New Learning



Hashing for Integrity, Passwords and Data Lookup; Hash input then Verify digest then Match purpose

I DO — WORKED EXAMPLE

Follow the worked method

A publisher provides a trusted SHA-256 digest of a file. Explain how a pupil can check the download.

Trusted published digest

Find the given data and name the required result.

Calculate downloaded hash

Explain why this intermediate stage is necessary.

Compare exact digests

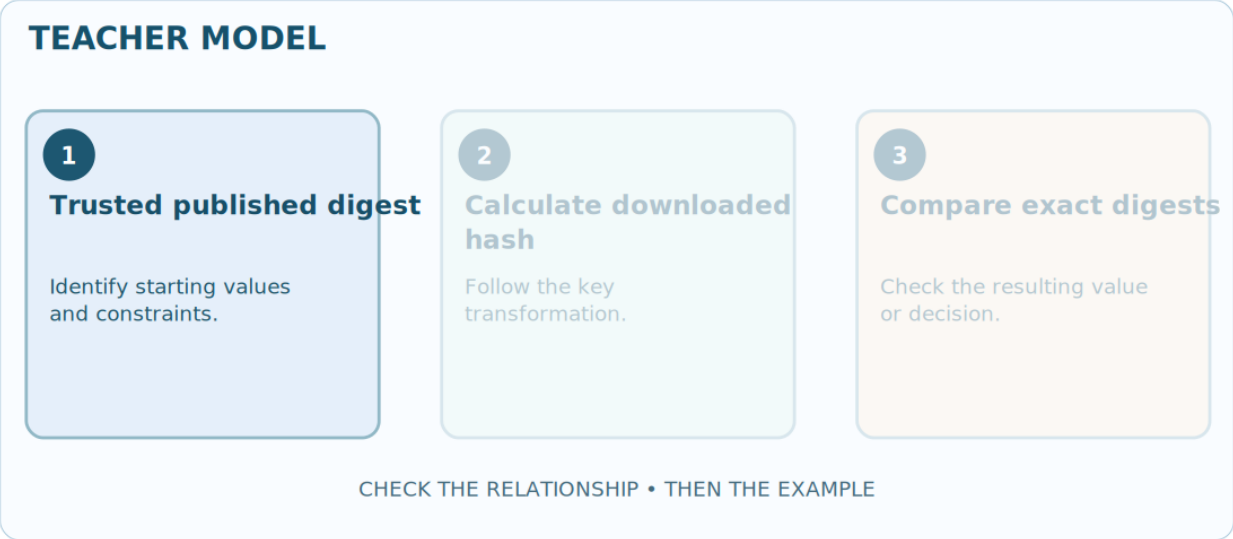
Calculate SHA-256 digest of received file and compare with the digest from a trusted source. If different, file changed or was corrupted. If equal, this strongly supports matching content, assuming trusted digest and secure algorithm; it does not by itself prove publisher identity.

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

A publisher provides a trusted SHA-256 digest of a file. Explain how a pupil can check the download.

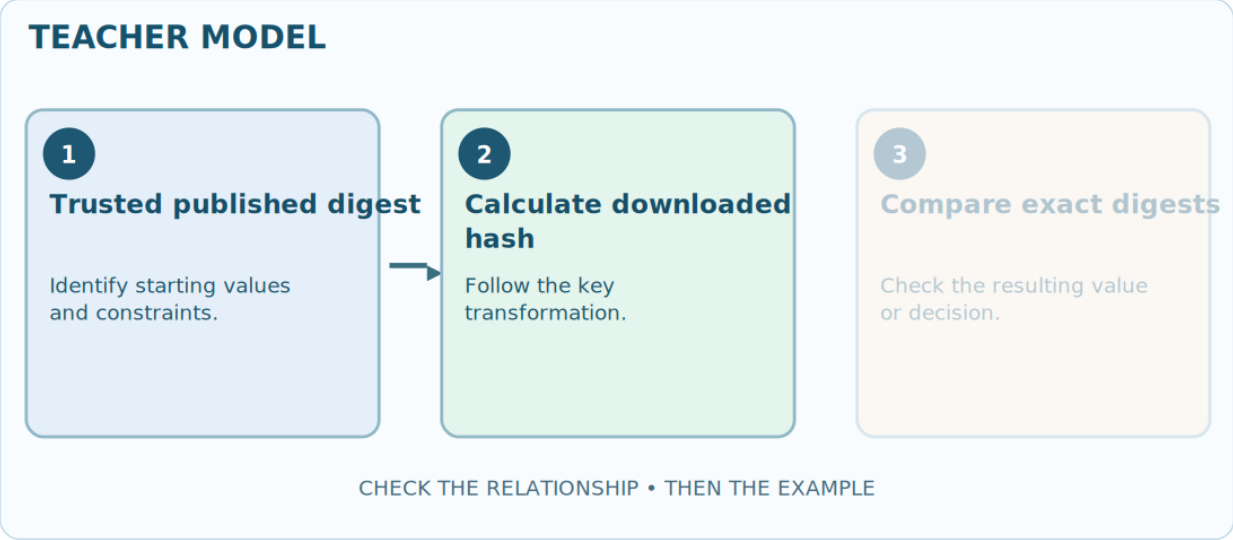
Find the given data and name the required result.



TEACHER MODEL; Trusted published digest then Calculate downloaded hash then Compare exact digests

A publisher provides a trusted SHA-256 digest of a file. Explain how a pupil can check the download.

Find the given data and name the required result.



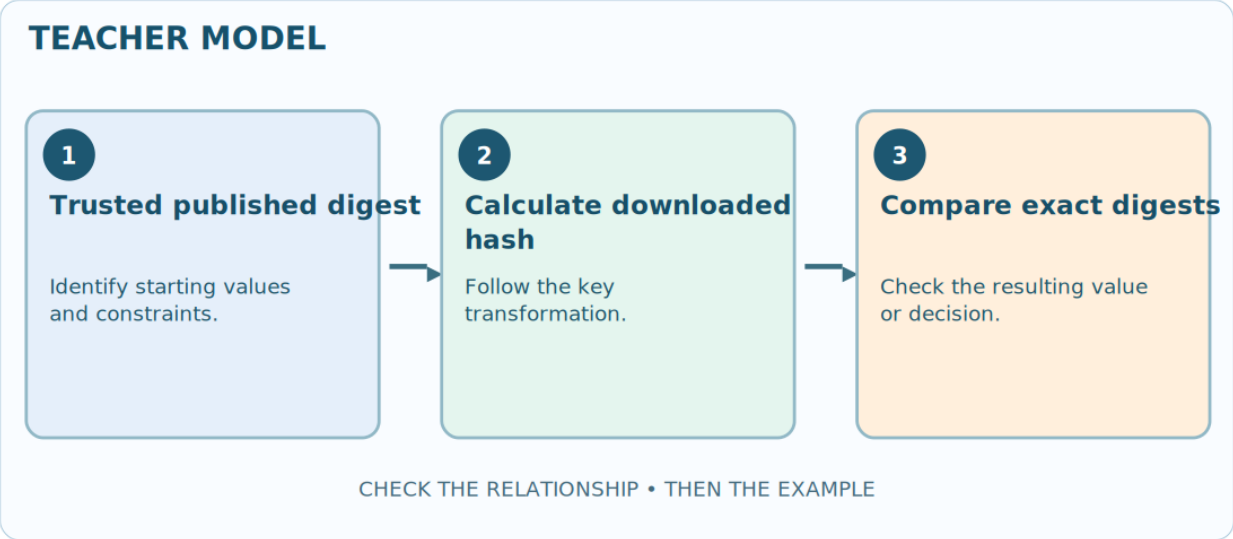
TEACHER MODEL; Trusted published digest then Calculate downloaded hash then Compare exact digests

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

A publisher provides a trusted SHA-256 digest of a file. Explain how a pupil can check the download.

Find the given data and name the required result.



TEACHER MODEL; Trusted published digest then Calculate downloaded hash then Compare exact digests

WE DO — GUIDED PRACTICE

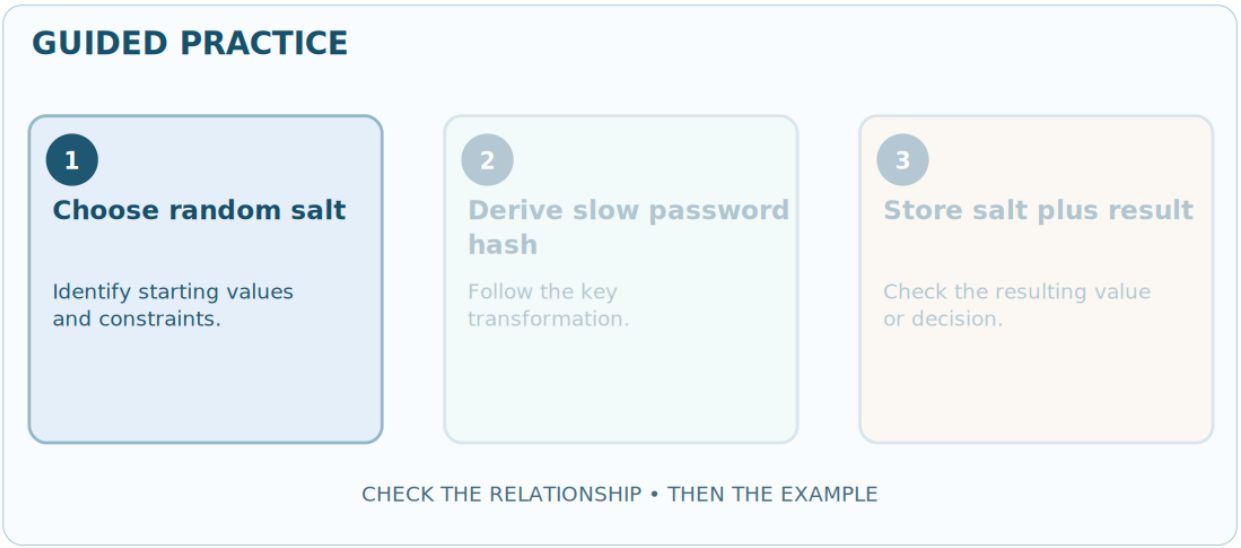
Try → Discuss → Improve

We Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

A database stores plaintext passwords. Explain a better use of hashing.

Find the given data and name the required result.



GUIDED PRACTICE; Choose random salt then Derive slow password hash then Store salt plus result

WE DO — TRY · DISCUSS · IMPROVE

TRY: Use the original guided questions and diagrams in the preceding We Do pages. DISCUSS: Explain and challenge methods before independent practice. IMPROVE: Note one change now; compare with the source model only AFTER the Exit Ticket.

Guided 1: Choose random salt Find the given data and name the required result.

DISCUSS — Compare your methods. Justify one important decision, identify one difference, and agree a specific improvement.

IMPROVE — Write one precise correction or stronger explanation in your own answer.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

State three different uses of hashing with one correct explanation each. [6]

Question 2

Explain what a hash collision is and why it matters. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Evaluate the claim 'a hash can decrypt a password and proves a file came from its claimed author'. [8]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

A hash output can normally be decrypted to recover the exact original password.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

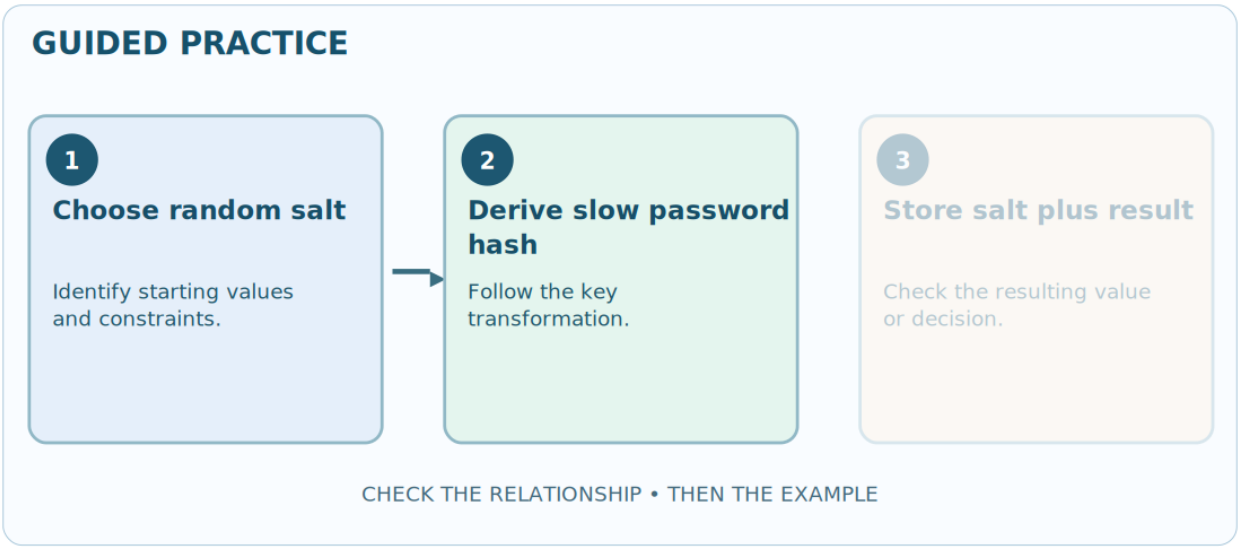
Next step: revisit the model or a relevant revision-guide section if needed.

We Do Check · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

A database stores plaintext passwords. Explain a better use of hashing.

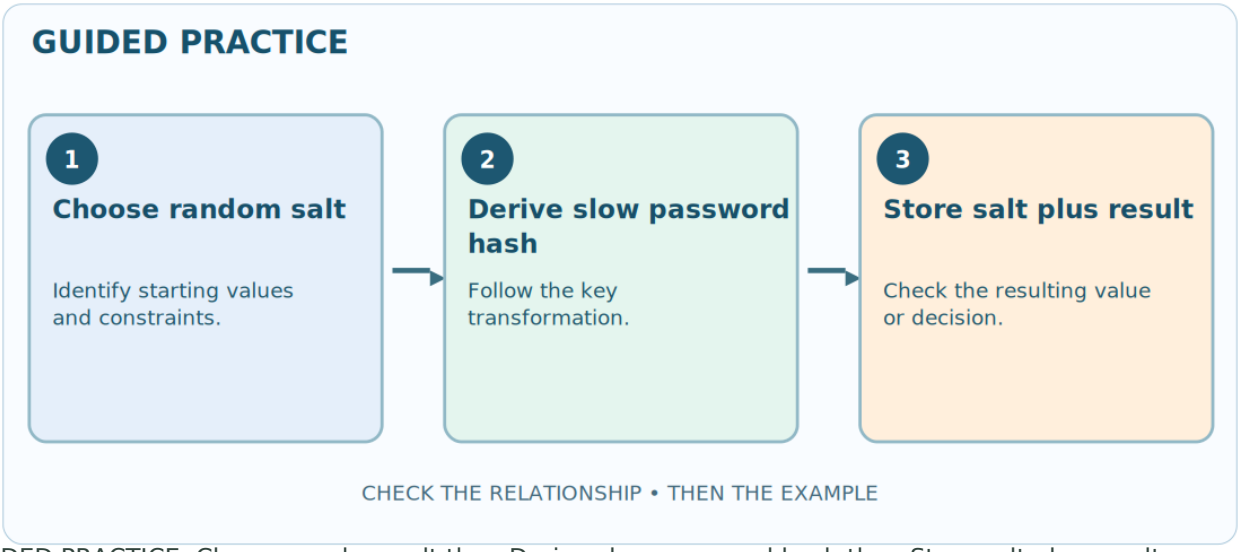
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GUIDED PRACTICE; Choose random salt then Derive slow password hash then Store salt plus result

A database stores plaintext passwords. Explain a better use of hashing.

Find the given data and name the required result.



GUIDED PRACTICE; Choose random salt then Derive slow password hash then Store salt plus result

WE DO — SOURCE CHECK / IMPROVE

Only use this section AFTER attempting the We Do, You Do and Exit Ticket. Compare with the original classroom lesson's CHECK explanation. Correct errors in GREEN PEN and explain why.

Guided 1 — original CHECK

Derive slow password hash

Explain why this intermediate stage is necessary.

Store salt plus result

Store unique salts and outputs from a slow password-hashing scheme such as Argon2id, not reversible encryption or a plain fast hash. On login, recompute using stored salt and compare securely.

IMPROVE: Explain one correction to your own reasoning.